

Technical Note:
Errors in the Analytic Photon Phase-Space Integrals of
MERADGEN 1.0 (**fsir.f**)
and Related Constants

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Abstract

Four integrals in the analytically integrated branch of the MERADGEN 1.0 real-photon cross section routine **fsir.f** are incorrect. The faults are a transcription error in **aj47**, two dimensional errors in **aj29**, a dimensional error and a sign error in **aj28**, and a sign error in **aj51**. Each fault is identified, located by line number, corrected, and verified analytically and numerically against the integral definitions given in the same code. All faulty terms carry an explicit factor of the electron mass squared and vanish in the ultrarelativistic limit, which is why comparison with the published ultrarelativistic results cannot detect them. The note also records two constants corrected in a C++ port used for Møller polarimetry and lists suspected misprints in the MERADGEN article.

1 Scope and context

MERADGEN 1.0 [1] generates radiative events in doubly polarized Møller scattering. Its routine **fsir.f** evaluates the real-photon emission cross section, fully differential (**ikey=0**), integrated over the photon variable z (**ikey=1**), or integrated over z and t_1 (**ikey=2**, and **ikey=-1** for the infrared-subtracted finite part). The ultrarelativistic limit of this calculation is published in Ref. [2]. The infrared divergence is treated with the covariant method of Ref. [4]; an exact lowest-order calculation without experimental cuts is given in Ref. [5]. The exact expressions beyond the ultrarelativistic limit are published only as this source code; Ref. [3] cites MERADGEN 1.0 as their reference implementation.

The issues were found while porting MERADGEN to C++ for use in Møller polarimetry for the MOLLER experiment at Jefferson Lab. Line numbers refer to the files **fsir.f** (857 lines) and **meradgen10.f** (860 lines) of the distributed code.

1.1 Structure of the calculation

fsir returns **coer*(sr1+...+sr10)**, line 847. Each **srN** is a linear combination of 51 basis integrals **aj1–aj51** with polynomial coefficients. In the **ikey=0** block (lines 43–96) each **ajNN** is defined as an integrand; the **ikey=1** block (lines 97–148) gives the z -integrated forms; the **ikey=2** block (lines 149–343) gives closed forms integrated over z and t_1 . The **ikey=2** closed forms are, by construction, the **ikey=0** integrands integrated over the photon phase space [1, Eqs. (13), (16)]. This definitional relation is the basis of every check below.

Notation follows the code: s, t, v are the Mandelstam variables and the inelasticity, $u = v - s - t + 4m^2$, $\lambda_s = s(s - 4m^2)$ (`als`), m^2 is the electron mass squared (`m2`), $z_2 = z - s - t_1 + 4m^2$, and $D = s + t_1 - 4m^2 = 1/dz$.

1.2 Summary of faults

Integral	Lines	Line changed	As distributed	Corrected
aj47	309–316	310	(v-v)	(v-u)
aj29	233–239	234	+8d0*m2)	+8d0*m2**2)
		238	/2d0/m2	/2d0/m2**2
aj28	230–232	230	((s*(v-t)-2d0*m2*v)*	((s*(v-t)-2d0*m2*v)*
		231	/2d0/m2	/2d0/m2**2
aj51	334–342	339	))+ (2d0*(	))- (2d0*(

Table 1: Faults in the `ikey=2` branch of `fsir.f`.

Integral	Definition (<code>ikey=0</code>)	Line	Used in
aj28	tt1*aj5	73	sr1 (358), sr5 (559)
aj29	tt1**2*aj5	74	sr1 (357)
aj47	m2*dz**3*aj8	92	sr10 (817)
aj51	dz**2*aj9	96	sr10 (811)

Table 2: Definitions and uses of the affected integrals.

2 Code faults

2.1 aj47: transcription error (line 310)

```
aj47=m2*(2d0*aj45*(s**2*(u-v)**2+6d0*m2**2*v**2-2d0*m2*s*(t*u
+2d0*v*(v-v)))-pi*(v*(-(s**3*u**2*v*(s**2-3d0*u**2+10d0*u*v
```

The factor $(v-v)$ is identically zero and should read $(v-u)$.

Justification. aj47 is the integral of $m^2/(D^3 z_2)$ over the photon phase space. After the z -integration (which reproduces the code’s own `ikey=1` form `aj8=-pi/sqrt(cz2)`) and the substitution $y = 1/D$, the t_1 -integral reduces exactly to a multiple of aj45 plus algebraic boundary terms. The multiple is fixed by the coefficients of `cz2` (from `zd`, `meradgen10.f` lines 815–834). Rewriting those coefficients with the kinematic identity $s + t + u = v + 4m^2$, i.e. $s + t - 4m^2 = v - u$, yields the coefficient of aj45 in the form printed in the code, with the term $2v(v - u)$ in place of $2v(v - v)$. Because aj45 is a logarithm and the boundary terms are algebraic, this coefficient is unique.

Numerical confirmation. With the correction, the closed form agrees with the numerical integral of the code’s own `ikey=1` expression over the code’s own t_1 limits (`meradgen10.f` lines 113–115) to better than 10^{-40} at four kinematic points. As distributed, it deviates by 0.4–1.2% at test points with m^2 comparable to s .

Structural cross-check. Under exchange of the two initial electrons at fixed detected electron, which is the $t \leftrightarrow u$ relation between the terms S_1 and S_{10} of Ref. [2], the **sr1** term $-4(2m^2 - s)^2 \mathbf{aj29}$ maps to the **sr10** group $-4(2m^2 - s)^2[\mathbf{aj51} + \mathbf{aj31} - 2\mathbf{aj47} + 2\mathbf{aj27}]$. The group arises from partial fractions of $m^2/(z^2 z_2^2)$; its four coefficients match the code exactly. The implied identity, group at (s, t, v) equal to **aj29** at (s, u, v) , holds to 10^{-39} with the corrections of this note and fails at the 10^{-4} – 10^{-7} level with **(v-v)**.

Magnitude. At beam energies of 1 and 11 GeV, $\theta_{\text{CM}} = 90^\circ$, and $v/(s + t)$ between 10^{-3} and 10^{-1} , the correction changes **aj47** by 6×10^{-8} to 7×10^{-5} relative.

2.2 aj29: dimensional errors (lines 234, 238)

```
.+v**2*(2d0*s*t*v*(s+2d0*m2)-v**2*((s-2d0*m2)**2+8d0*m2)
...
.dlog((s-2d0*m2)*(s-2d0*m2+dsqrt(als))/2d0/m2
.-1d0))
```

Line 234. $(s - 2m^2)^2$ has mass dimension four; **8d0*m2** has dimension two. The correct term is **8d0*m2**2**.

Line 238. The argument of a logarithm must be dimensionless. The product $(s - 2m^2)(s - 2m^2 + \sqrt{\lambda_s})$ has dimension four, so it must be divided by m^4 , not m^2 . With m^4 , the argument is the square of a dimensionless ratio and the logarithm is

$$\ell_s = 2 \ln \frac{s - 2m^2 + \sqrt{\lambda_s}}{2m^2} = 2 \ln \frac{1 + \xi_s}{1 - \xi_s}, \quad \xi_s = \frac{\sqrt{s - 4m^2}}{\sqrt{s}}. \quad (1)$$

This is the logarithm used, without approximation, in Ref. [3, Eqs. (47), (54), (60)]. Its ultrarelativistic limit is $\ln(s^2/m^4) = s L_s$ with L_s of Ref. [2, Eq. (A11)]. The same logarithm is coded correctly in **aj21** (line 204, divided by **m2**2**) and, in simplified form, in **aj36** (line 268). As distributed, the argument depends on the unit system: in GeV the logarithm is shifted by $\ln m^2 \approx -15.2$.

Both errors are invisible when $m^2 = 1$ in the units used.

(s, t, v, m^2)	As distributed	Corrected
(20, -5, 1, 1)	1×10^{-41}	1×10^{-41}
(10, -3, 0.5, 0.2)	-4.8×10^{-3}	2×10^{-41}
(8, -2, 0.7, 0.3)	-1.6×10^{-2}	2×10^{-41}
(1, -0.4, 0.05, 10^{-3})	-8.7×10^{-4}	3×10^{-41}

Table 3: Relative deviation of the **aj29** closed form from the integrated **ikay=1** expression. Only the combination of both corrections reaches agreement.

Numerical confirmation.

2.3 aj28: dimensional error and sign error (lines 230, 231)

```

aj28=m2*pi/t**2/als*((s*(v-t)-2d0*m2*v)*
.dlog((s-2d0*m2)*(s-2d0*m2+dsqrt(als))/2d0/m2
.-1d0)/dsqrt(als)-(t*(s*v-als)-2d0*m2*v**2)/m2/v)

```

Line 231. The logarithm is the same ℓ_s as in aj29, with the same missing power of m2. aj28 and aj29 are both built on aj5 (lines 73–74), and the identical slip suggests one expression was copied from the other.

Line 230. Correcting the logarithm alone does not reconcile aj28; the logarithmic term also has the wrong sign. Exact integration was carried out by the same reduction used for aj51 (Sec. 2.4), with $y = 1/t_1$. Separating the exact result into its algebraic and logarithmic parts and comparing each with the code gives a ratio of exactly +1 for the algebraic part and exactly –1 for the logarithmic part at every point tested. The error therefore is not in the structure of the expression but in the sign of one term.

Numerical confirmation. As distributed, aj28 deviates from the integrated `key=1` expression by –5.0%, –1.9%, –6.9%, and –0.12% at the four points of the aj29 table; it fails even at $m^2 = 1$, where the logarithm correction has no effect. With both corrections, agreement is better than 3×10^{-51} at five points.

Scope. aj28 also enters sr5 (line 559), a t – u interference term.

2.4 aj51: sign error (line 339)

```

--s*u*v)))+(2d0*(-3d0*(m2**2+(s-m2)**2)*u*v**2+3d0*m2*(s-2d0*m2)

```

The sign preceding the logarithmic term should be negative.

Analytic justification. aj51 is the integral of $m^2/(D^2 z_2^2)$; aj45 is the integral of $1/(D z_2)$ over the same region. The derivation uses only definitions contained in the code.

1. The z -integral of $1/z_2$ is $-\pi/\sqrt{c_{z2}}$ with $c_{z2} = a_z D^2 + 2b_z D + c_z$. Differentiating with respect to D gives the z -integral of $1/z_2^2$ as $\pi b_{z2}/c_{z2}^{3/2}$, using the exact identity $b_{z2} = a_z D + b_z$. This reproduces the code's `key=1` form `aj9=m2*pi*bz2/cz2**1.5`.
2. Writing $c_{z2} = \alpha D^2 + \beta D + \gamma$, $b_{z2} = pD + q$, and substituting $y = 1/D$ gives

$$\text{aj45} = \pi \int \frac{dy}{\sqrt{Q}}, \quad \text{aj51} = -\pi m^2 \int \frac{y^2(p + qy)}{Q^{3/2}} dy, \quad Q = \gamma y^2 + \beta y + \alpha, \quad (2)$$

with identical limits and orientation.

3. The second integral reduces exactly to an algebraic term plus $c \int dy/\sqrt{Q}$, where the constant c follows from a linear system and equals $2K/(\lambda_s^2 u^3)$, with K the polynomial that multiplies the logarithm in the code. Hence

$$\text{aj51} = -\frac{2m^2 K}{\lambda_s^2 u^3} \text{aj45} + \text{algebraic terms.} \quad (3)$$

4. The code's logarithmic term, rewritten with its own `aj45` (line 305), equals $+2m^2 K \text{aj45}/(\lambda_s^2 u^3)$: correct magnitude, opposite sign.
5. The coefficient of `aj45` is unique because a logarithm cannot equal an algebraic function.

(s, t, v, m^2)	As distributed	Sign corrected
(20, -5, 1, 1)	-0.70	-2×10^{-51}
(10, -3, 0.5, 0.2)	-0.53	-3×10^{-51}
(8, -2, 0.7, 0.3)	-0.46	4×10^{-51}
(1, -0.4, 0.05, 10^{-3})	-0.12	2×10^{-51}

Table 4: Relative deviation of the `aj51` closed form from the integrated `ikkey=1` expression.

Numerical confirmation. Separating the exact integral into parts gives a ratio of exactly +1 for the algebraic part and exactly -1 for the logarithmic part. As $m^2 \rightarrow 0$ at fixed s, t, v , the absolute error falls roughly as m^2 , from 8×10^{-2} at $m^2 = 10^{-3}$ to 3×10^{-7} at $m^2 = 10^{-9}$.

2.5 Why the faults were not detected

- Every faulty term carries an explicit factor of m^2 and vanishes in the ultrarelativistic limit. In that limit `sr1` and `sr10` reproduce S_1 and S_{10} of Ref. [2] to 10^{-30} relative (checked at $m^2 = 10^{-30}$, $P = 0, \pm 1$), so comparison with the published formulas cannot see these errors.
- The two dimensional errors are invisible whenever $m^2 = 1$ in the chosen units.
- The built-in tests of Ref. [1, Sec. 6] compare generated distributions with cross sections computed by the same routine and are insensitive to errors in the routine itself.

3 Verification methods and cited inputs

Inputs from the code. The analytic arguments use only: the `ikkey=0` definitions (`fsir.f` lines 43–96); the z -quadratic coefficients from `zd` (`meradgen10.f` line 815; a_z at 821, b_{z2} at 832, c_{z2} at 834), together with the exact identities $c_{z2} = a_z D^2 + 2b_z D + c_z$ and $b_{z2} = a_z D + b_z$; the t_1 limits (`meradgen10.f` lines 113–115); and λ_s (`als`, `meradgen10.f` line 272).

Methods.

1. Self-consistency: numerical integration of the code's own `ikkey=1` expressions over the code's own t_1 limits, compared with the `ikkey=2` closed forms.
2. Independent integration: direct two-dimensional integration of the `ikkey=0` integrands over the physical region defined by the Gram determinant $\Delta(k_1, k_2, p_1, k)$.
3. Exact antiderivatives by reduction to $\int dy/\sqrt{Q}$, validated against methods 1 and 2 to 10^{-49} .
4. Ultrarelativistic comparison of `sr1` and `sr10` with S_1 and S_{10} of Ref. [2], using the logarithms of its Eq. (A11).

Arithmetic used 30 to 80 significant digits (Python `mpmath`); algebraic identities were checked with `SymPy`. Test points use m^2 comparable to s deliberately, to expose mass-suppressed terms.

Coverage. Fifteen of the 51 integrals have been checked: aj1, aj5, aj8, aj9, aj27, aj28, aj29, aj31, aj45–aj51. The remaining 36 have not.

4 Items corrected in the C++ port

Electron mass squared (meradgen10.f line 268). $m2/.261112d-6/$ is inconsistent with $m/.511000d-3/$, whose square is 0.261121×10^{-6} ; the last two digits are transposed. Since s is built from both m and $m2$, the kinematics are slightly inconsistent. The port computes m^2 from m and uses the current PDG value [7].

Lepton masses in vacpol (meradgen10.f line 366). Updated per PDG [7]. As distributed, the electron entry ($.26110d-6$) differs from line 268, so the code carries two different electron masses, and the τ entry corresponds to $m_\tau \approx 1.784$ GeV, a pre-1992 value.

Hadronic vacuum polarization (following line 366). Retained as distributed (parametrization of Ref. [6]). At polarimeter kinematics, $Q^2 \approx 5.6 \times 10^{-3}$ GeV² at 11 GeV and $\theta_{CM} = 90^\circ$, the contribution is of order 10^{-4} in the cross section and largely cancels in the analyzing power.

5 Suspected misprints in Ref. [1]

These items concern the article only; the code in `dcanc` (meradgen10.f lines 443–467) is consistent with Ref. [2, Eq. (29)] and Ref. [3, Eq. (61)]. They were identified from a text-extracted copy and should be confirmed against the typeset article.

- Eq. (B.10), δ_1^H : the bracket contains $Li_2(v_{\max}/t)$ twice; the second should be $Li_2(v_{\max}/(s+t))$, as in `dcanc` line 462.
- Eq. (B.10), δ_1^H : $Li_2((s - v_{\max})/s)$ appears with a negative sign; the code and Refs. [2, 3] have a positive sign.

6 Recommendations

1. Apply the corrections of Table 1 in any use of `fsir` beyond the ultrarelativistic limit.
2. Check the remaining 36 integrals with method 1 of Sec. 3, which requires only the distributed code.
3. Add a unit-scaling test: scaling s , t , v , m^2 by a common factor κ^2 must scale each integral by a fixed power of κ . This test detects dimensional errors such as those in aj28 and aj29.
4. Test with $m^2 \neq 1$ and with m^2 not small compared with s .
5. Quantify the combined effect of the corrections on the cross section and analyzing power at polarimeter kinematics; this has not yet been done.
6. Report the findings to the authors [1] and request confirmation against their original derivation.

References

- [1] A. Afanasev, E. Chudakov, A. Ilyichev, and V. Zykunov, “MERADGEN 1.0: Monte Carlo generator for the simulation of radiative events in parity conserving doubly-polarized Møller scattering,” *Comput. Phys. Commun.* **176**, 218 (2007), arXiv:hep-ph/0603027.
- [2] A. Ilyichev and V. Zykunov, “Lowest order QED radiative corrections to longitudinally polarized Møller scattering,” *Phys. Rev. D* **72**, 033018 (2005).
- [3] I. Akushevich, H. Gao, A. Ilyichev, and M. Meziane, “Radiative corrections beyond the ultra relativistic limit in unpolarized ep elastic and Møller scatterings for the PRad experiment at Jefferson Laboratory,” *Eur. Phys. J. A* **51**, 1 (2015).
- [4] D. Yu. Bardin and N. M. Shumeiko, *Nucl. Phys.* **B127**, 242 (1977).
- [5] N. M. Shumeiko and J. G. Suarez, *J. Phys. G* **26**, 113 (2000).
- [6] H. Burkhardt and B. Pietrzyk, *Phys. Lett. B* **356**, 398 (1995).
- [7] S. Navas *et al.* (Particle Data Group), “Review of Particle Physics,” *Phys. Rev. D* **110**, 030001 (2024).